Business Problem

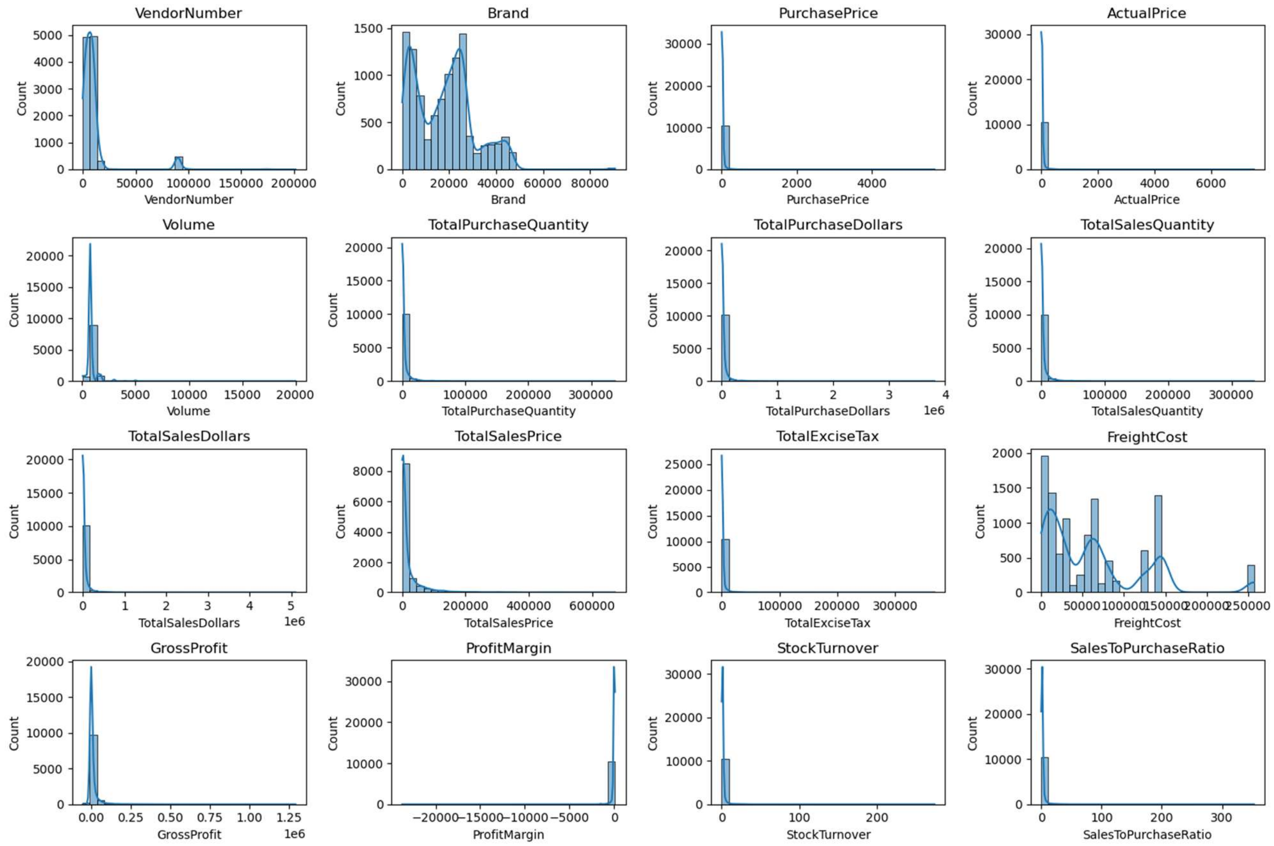
Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.360000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-15.620770	443.555329	-23730.638953	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02



Negative & Zero Values

Gross Profit:
The presence of negative Gross Profit values indicates that some products were sold at a loss, where the total sales revenue was lower than the total purchase cost. This may result from heavy discounts, damaged goods, pricing errors, or clearance sales.

Profit Margin:
Profit Margin contains -∞ (negative infinity) values, which typically occur when the denominator (Sales Revenue) is zero while Gross Profit is negative. This indicates products with purchase costs but no recorded sales, making the profit margin mathematically undefined.

Total Sales Quantity & Total Sales Dollars:
Some products have zero sales quantity and zero sales revenue, indicating purchased inventory that has not yet been sold. These products may represent slow-moving, obsolete, or newly stocked inventory.

Outliers Detected

Purchase Price & Actual Price:
Both variables exhibit strong right-skewed distributions with a few extremely high-priced products compared to the majority. These high-value observations likely correspond to premium or luxury products and should be investigated separately if they influence the analysis.

Freight Cost:

Freight Cost shows substantial variability with several extreme outliers. This may be caused by large bulk shipments, long-distance transportation, special handling requirements, or inconsistent shipping practices. These outliers can significantly affect average freight cost calculations.

Total Purchase Quantity & Total Sales Quantity:

Both variables are highly right-skewed, indicating that while most products are purchased and sold in relatively small quantities, a few products have exceptionally large transaction volumes. These high-volume products may dominate inventory and sales analysis.

Total Purchase Dollars & Total Sales Dollars:

The distributions contain a small number of very high-value transactions compared to the majority of products, suggesting that a limited number of products contribute disproportionately to total revenue and procurement costs.

Stock Turnover:
Stock Turnover ranges from 0 to very high values, indicating considerable variation in inventory movement. A turnover of 0 represents products with no sales, while values greater than 1 indicate inventory is replenished and sold multiple times during the analysis period. Extremely high turnover values may correspond to fast-moving products or frequent restocking.

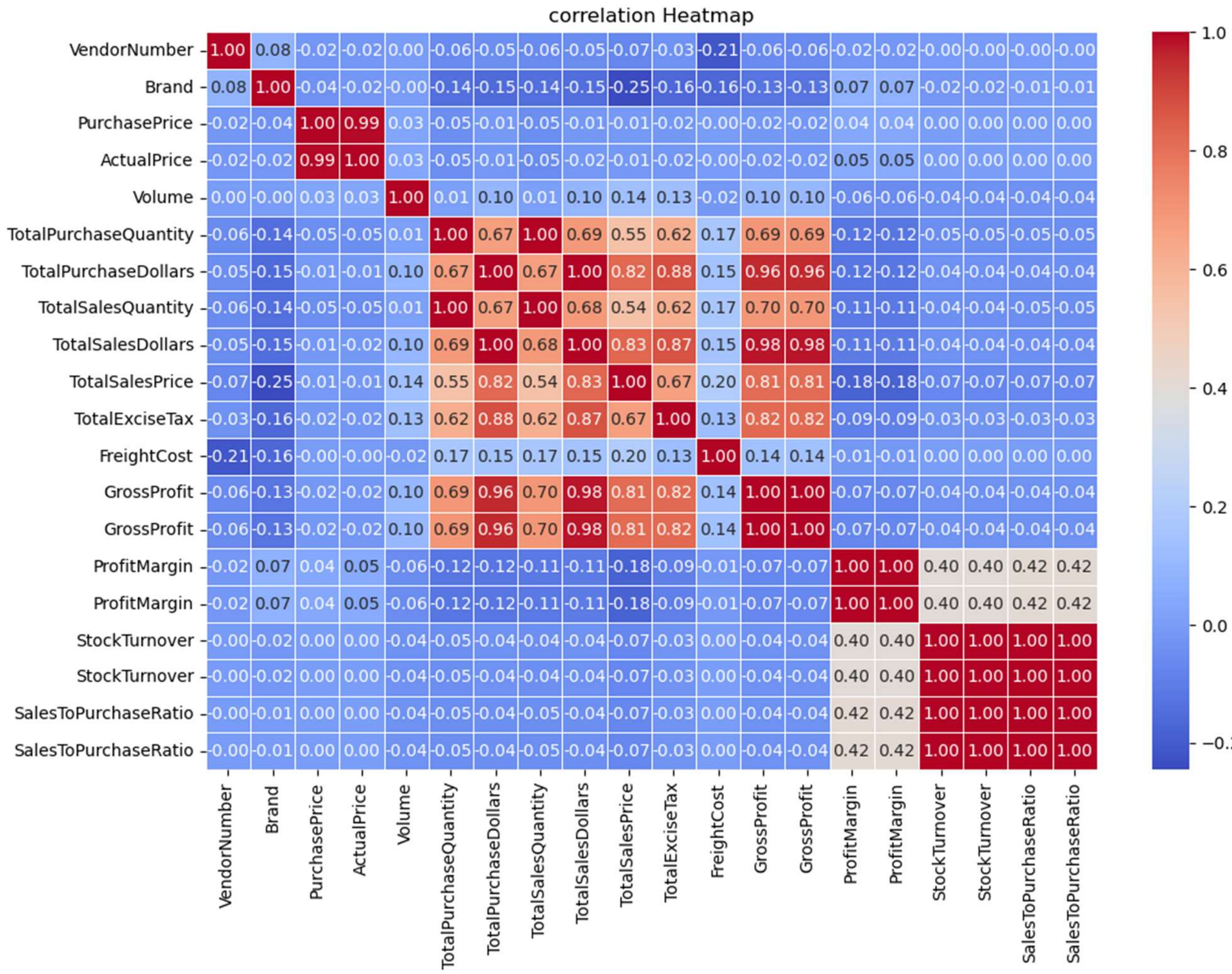
Sales-to-Purchase Ratio:
The Sales-to-Purchase Ratio is highly right-skewed. Values greater than 1 indicate that sales exceed purchases during the observed period, often because sales are fulfilled using inventory purchased in previous periods rather than current purchases.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Correlation Insights



Purchase Price vs. Total Sales Dollars & Gross Profit:
Purchase Price has almost no correlation with Total Sales Dollars (-0.01) and Gross Profit (-0.02), indicating that individual product purchase prices have little influence on the overall sales revenue or gross profit. Business performance appears to be driven more by sales volume than unit price.

Total Purchase Quantity vs. Total Sales Quantity:
Total Purchase Quantity and Total Sales Quantity exhibit an almost perfect positive correlation (1.00), indicating that products purchased in higher quantities are generally sold in proportionally higher quantities. This suggests efficient inventory planning and demand forecasting.

Profit Margin vs. Total Sales Price:
Profit Margin has a weak negative correlation (-0.18) with Total Sales Price, suggesting that products generating higher total sales revenue do not necessarily achieve higher profit margins. This may be due to discounting, competitive pricing, or higher associated costs.

Stock Turnover vs. Gross Profit & Profit Margin:
Stock Turnover shows very weak negative correlation with Gross Profit (-0.04) and a moderate positive correlation with Profit Margin (0.40). This indicates that increasing inventory turnover has little direct impact on gross profit but is moderately associated with improved profit margins.

Research Questions & Key Findings

1. Brands for Promotional or Pricing Adjustments

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Brands with Low Sales but High Profit Margins:
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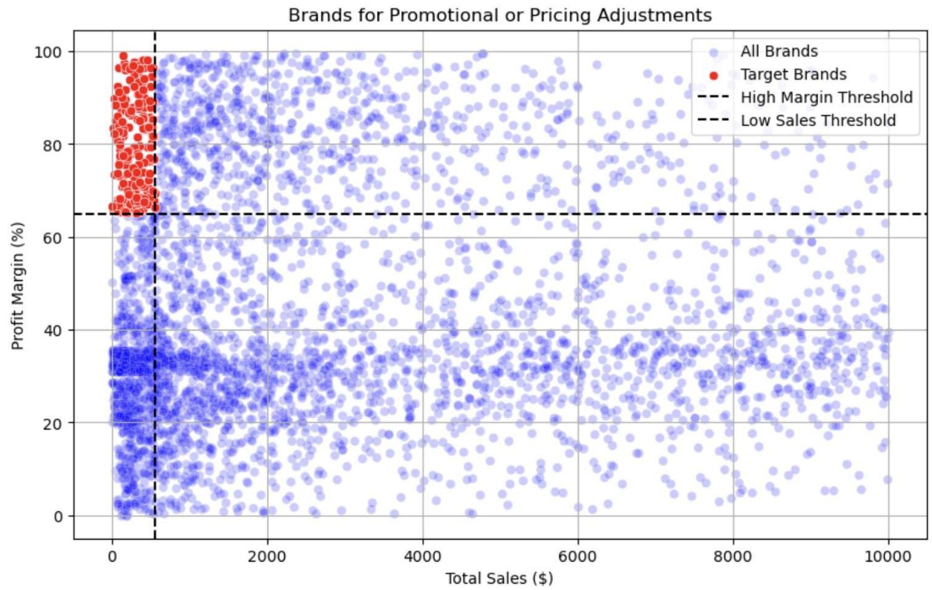
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

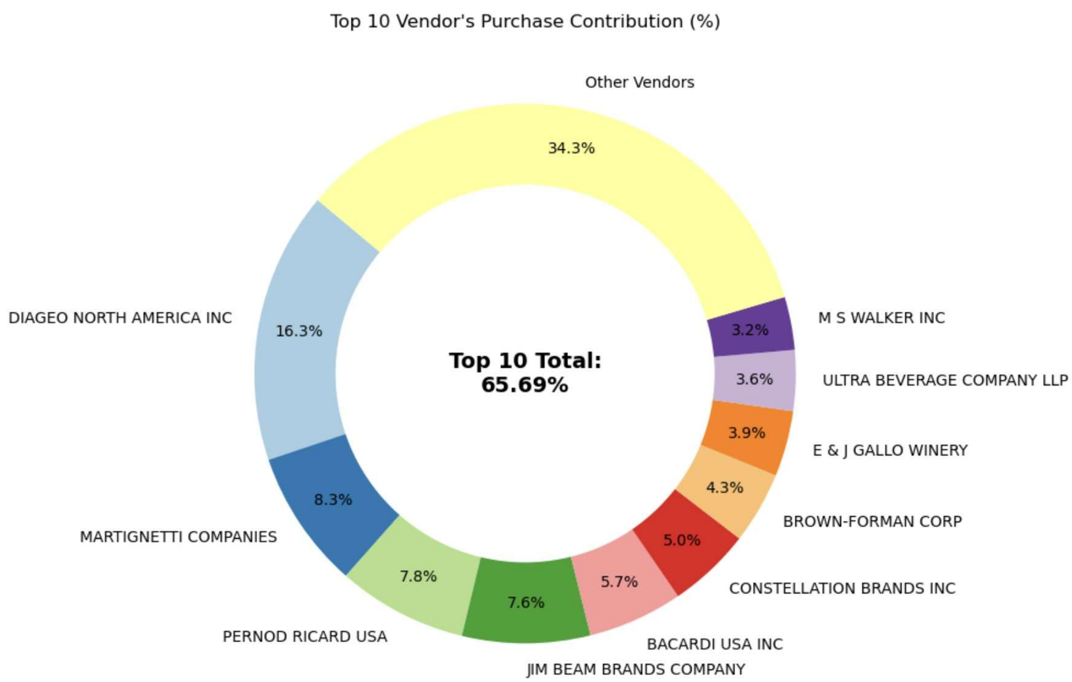
198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.



2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost ($10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

OrderSize	UnitPurchasePrice
Small	39.057543
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: $2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

VendorName	UnsoldInventoryValue
DIAGEO NORTH AMERICA INC	722.21K
JIM BEAM BRANDS COMPANY	554.67K
PERNOD RICARD USA	470.63K
WILLIAM GRANT & SONS INC	401.96K
E & J GALLO WINERY	228.28K
SAZERAC CO INC	198.44K
BROWN-FORMAN CORP	177.73K
CONSTELLATION BRANDS INC	133.62K
MOET HENNESSY USA INC	126.48K
REMY COINTREAU USA INC	118.60K

5. Profit Margin Comparison: High vs. Low-Performing Vendors

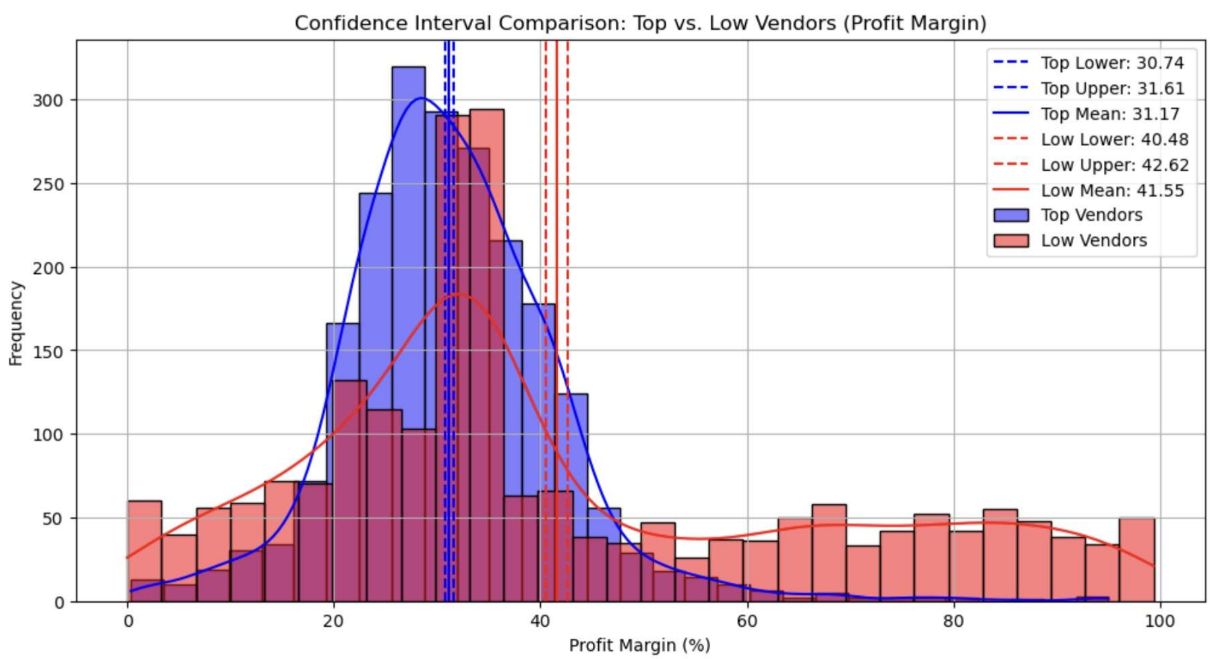
Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%

Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

H_0 (Null Hypothesis): No significant difference in profit margins between top and low-performing vendors.

H_1 (Alternative Hypothesis): A significant difference exists in profit margins between the two vendor groups.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

Final Recommendations

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.